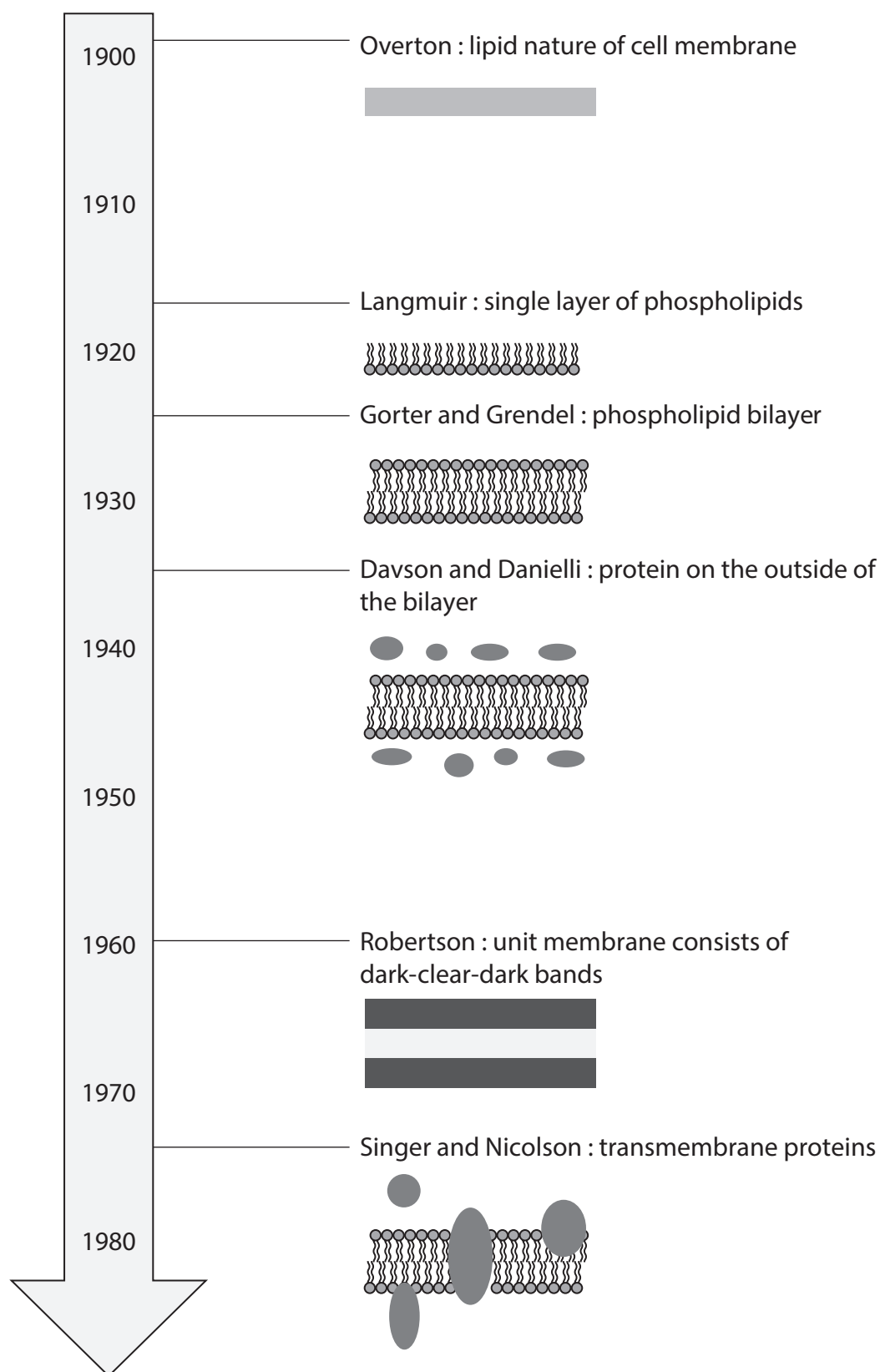


- 7 The structure of the cell membrane has been modelled to interpret the data available at the time.

The diagram shows a timeline of six models proposed for the structure of the cell membrane.



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- (a) Overton suggested that some types of solute could cross the lipid membrane, whereas other types of solute could not.

Explain the difference between these two types of solute and why only one type can cross the lipid membrane.

(2)

- (b) Langmuir's model was a single layer of phospholipids.

Explain why the phospholipids in a cell membrane cannot be arranged in a single layer.

(2)

- (c) Gorter and Grendel suggested that there were various types of phospholipid but they were all arranged in a bilayer.

Give **two** ways in which phospholipids can be different from each other.

(2)

1

2

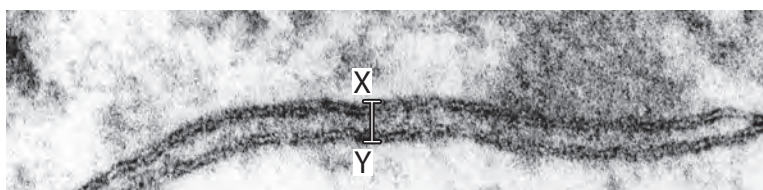
- (d) The Davson and Danielli model was the first to include proteins.

Explain why some membrane proteins would **not** be able to carry out their transport functions if they were located only on the surface of the phospholipid bilayer.

(2)

- (e) Robertson implied that there was no space between the phospholipid layers and that the thickness of the membrane could be measured.

The photograph shows part of two adjacent cell membranes.



(Source: © DENNIS KUNKEL MICROSCOPY/SCIENCE PHOTO LIBRARY)

The thickness of these cell membranes between X and Y is 22.8 nm.

Calculate the magnification of this photograph.

Express your answer to **two** significant figures.

(2)

Answer



- (f) In 1972, Singer and Nicolson proposed the fluid mosaic model of the cell membrane.

Explain why their model has been given this name.

(2)

(Total for Question 7 = 12 marks)

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